# UGANDA INDUSTRIAL RESEARCH INSTITUTE

Namanve Industrial and Business Park, Mukono District, Uganda

## SOFTWARE REQUIREMENTS SPECIFICATION

*FOR THE DESIGN AND DEVELOPMENT OF A WEB BASED*

*INVENTORY MANAGEMENT SYSTEM*

*FOR UGANDA INDUSTRIAL RESEARCH INSTITUTE (UIRI)*

*NAKAWA AND NAMANVE CAMPUSES*

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# Document – Change Control

This document forms the Software Requirements Specification for the UIRI Inventory Management System. It translates the approved proposal and project brief into testable system requirements for database design, system design, implementation and validation.

| Item | Description |
| --- | --- |
| **Document Title** | Software Requirements Specification for the UIRI Inventory Management System |
| **Project Type** | Database and Web Application Development |
| **Primary User Department** | Information and Communications Technology (ICT) |
| **Initial Pilot Site** | UIRI Namanve |
| **Planned Institutional Coverage** | Nakawa and Namanve Campuses |
| **Prepared for** | ICT Team, Uganda Industrial Research Institute |
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*Table 1 Document Control Details*

## Revision History

| Version | Date | Description | Author |
| --- | --- | --- | --- |
| **1.0** | July-August 2026 | Initial SRS prepared after completion of the system proposal | Keith |
| **2.0** | TBD | TBD | TBD |
| **3.0** | TBD | TBD | TBD |

*Table 2 Revision History*

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## Abbreviations and Acronyms

| Acronym | Full Form |
|---|---|
| **API** | Application Programming Interface |
| **DBMS** | Database Management System |
| **ERD** | Entity Relationship Diagram |
| **FR** | Functional Requirement |
| **ICT** | Information and Communications Technology |
| **IMS** | Inventory Management System |
| **MMISDC** | Machining, Manufacturing and Industrial Skills Development Centre |
| **NFR** | Non-Functional Requirement |
| **PDU** | Procurement and Disposal Unit |
| **RBAC** | Role Based Access Control |
| **SRS** | Software Requirements Specification |
| **SQL** | Structured Query Language |
| **UIRI** | Uganda Industrial Research Institute |

*Table 3 Acronyms and Abbreviations*

# 1.0 Introduction

## 1.1 Purpose

This Software Requirements Specification defines the functional and non functional requirements for the proposed web based Inventory Management System for Uganda Industrial Research Institute. The document converts the system proposal and the UIRI ICT Team project brief into clear, testable and traceable requirements for the development team, supervisor, assessors and future system users [1], [2].

The SRS will guide the next project stages, including data dictionary preparation, conceptual and logical ERD design, normalization, UML diagrams, source code implementation, testing, user manual preparation and final demonstration.

## 1.2 Scope of the System

The system shall provide a centralized database driven platform for registering, categorizing, tracking and reporting inventory records across UIRI. It shall support inventory item management, individual asset tracking, stock receipt, stock issue, stock transfer, stock return, supplier management, user authentication, role based access control, low stock alerts, dashboard reporting and audit logging.

The database shall support both Nakawa and Namanve campuses. The initial implementation may be piloted at UIRI Namanve because the internship project supervision is based there, while the data model shall remain capable of supporting Nakawa without redesign.

## 1.3 Intended Audience

This document is intended for the UIRI ICT Team, the field supervisor, academic supervisors, project assessors, student developers, store managers, departmental staff and any future technical team that may maintain or extend the system.

## 1.4 Document Conventions

Functional requirements are labelled using the format FR-MODULE-NN. Non functional requirements are labelled using the format NFR-CATEGORY-NN. The word shall indicates a mandatory requirement. The word should indicates a desirable requirement that may be implemented where time and resources allow.

## 1.5 References

| Reference | Document or Source |
|---|---|
| **[1]** | ICT Team, UIRI Namanve, Internship Project Question, 2026 |
| **[2]** | Project Team, System Proposal for the UIRI Inventory Management System, 2026 |
| **[3]** | Uganda Industrial Research Institute Official Website |
| **[4]** | Ministry of Trade, Industry and Cooperatives, UIRI profile |
| **[5]** | T. M. Connolly and C. E. Begg, Database Systems, 4th ed. |
| **[6]** | IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications |

# 2.0 Overall Description

## 2.1 Product Perspective

The proposed Inventory Management System shall be a standalone browser accessible application connected to a relational database. It is intended to replace scattered manual records, separate spreadsheets and slow reporting procedures with one controlled system for inventory visibility and accountability.

The system shall use a three layer arrangement consisting of a user interface layer, an application logic layer and a data layer. The user interface shall run in a web browser. The application logic shall process requests, enforce validation and apply access control. The data layer shall store normalized MySQL or MariaDB tables.

## 2.2 Product Functions

The main system functions shall include login and access control, user and role management, department and campus setup, item category management, inventory item registration, individual asset registration, stock receiving, stock issuing, stock transfer, stock return, supplier management, condition tracking, low stock alerts, dashboard reporting, search, filtering and audit logging.

## 2.3 User Classes and Access Summary

| User Class | Typical User | Main Access Rights |
|---|---|---|
| **Administrator** | ICT staff or authorized system controller | Manage users, roles, campuses, departments, categories, settings, reports and audit logs |
| **Store Manager** | Stores or inventory custodian | Register items, receive stock, issue stock, update suppliers, view stock balances and generate inventory reports |
| **Staff User** | Departmental user from ICT, Research, Administration, Finance, HR, MMISDC or CCTV | Search authorized inventory records, view availability and make or confirm inventory related requests |
| **Management Viewer** | Supervisor, management or authorized reviewer | View dashboard summaries, department reports, stock movement reports and asset status reports |
| **Auditor** | Internal reviewer where permitted | Review audit logs, transaction history and inventory accountability reports |

*Table 5 User Classes and Access Summary*

## 2.4 Operating Environment

| Area | Requirement |
|---|---|
| **Server Environment** | XAMPP or equivalent local development stack with Apache, PHP and MySQL or MariaDB |
| **Client Environment** | Modern web browser running on desktop or laptop computers within the local network |
| **Database Environment** | MySQL or MariaDB relational database with primary keys, foreign keys, constraints and indexes |
| **Deployment Context** | Initial local intranet deployment at UIRI Namanve with future extension to Nakawa |
| **Development Tools** | PHP, HTML, CSS, Bootstrap, JavaScript, Chart.js, phpMyAdmin and Microsoft Visio for design diagrams |

*Table 6 Operating Environment*

## 2.5 Assumptions and Dependencies

- The ICT Team will provide guidance on final user roles and approved workflows.
- Real inventory data will only be used where permission is granted by authorized personnel.
- Where real data is incomplete or sensitive, dummy data will be used for demonstration and testing.
- The hosting machine or local server will be available during development and demonstration.
- Historical inventory records may require cleaning before they can be entered into the database.

## 2.6 Design Constraints

- The database shall be normalized to at least Third Normal Form to reduce duplication and preserve data consistency [5].
- The system shall use role based access control so that users only access functions allowed by their role.
- The system shall use input validation and parameterized database queries to reduce SQL injection risks.
- The initial version shall not include barcode scanning, RFID or IoT telemetry unless approved as a future enhancement.

# 3.0 External Interface Requirements

## 3.1 User Interface Requirements

The user interface shall be simple enough for non technical staff to use during daily inventory operations. Forms shall contain clear labels, required field indicators, validation messages and success or failure feedback. Tables shall support readable columns for item code, item name, category, campus, department, quantity, condition and date.

- The login page shall request username or email and password.
- The dashboard shall display summary cards for total items, available stock, low stock items, faulty assets and recent transactions.
- The inventory list shall allow search and filtering by name, code, category, supplier, campus, department, condition and date range.
- Reports shall be printable or exportable where practical during the prototype stage.

## 3.2 Hardware Interface Requirements

No specialized client hardware shall be required for the initial version. Users shall access the system through ordinary desktop or laptop computers. Barcode readers, RFID readers and IoT sensors are excluded from the initial scope and may be considered after the core database has been tested.

## 3.3 Software Interface Requirements

The system shall interface with a MySQL or MariaDB database. During development, phpMyAdmin may be used for database administration. The system may use Chart.js for dashboard charts and Bootstrap for responsive interface layout.

## 3.4 Communication Interface Requirements

The initial deployment shall operate over the local intranet. User sessions shall be managed securely on the server side. The system shall not expose inventory records to external systems unless authorized by UIRI in a future integration phase.

# 4.0 Functional Requirements

This section defines the mandatory functions of the system. Each requirement is written in a testable form and grouped according to the module it belongs to.

## 4.1 Functional Requirements Summary

| Module | Purpose |
|---|---|
| **Authentication and Authorization** | Login, logout, password protection, user roles and access permissions |
| **User and Master Data Management** | Users, roles, campuses, departments, locations and categories |
| **Inventory Catalogue and Asset Tracking** | Item records, asset instances, serial numbers, brand, purchase year, batch and condition |
| **Stock Management** | Stock received, issued, transferred, returned, adjusted and balanced |
| **Supplier Management** | Supplier records and supplier transaction history |
| **Reports and Dashboard** | Low stock, stock movement, asset condition, department inventory and periodic reports |
| **Security and Audit Trail** | Input validation, access restriction, password hashing and user activity logging |

*Table 7 Functional Requirements Summary*

## 4.2 Authentication and Authorization Requirements

| Requirement ID | Requirement Statement | Priority |
|---|---|---|
| **FR-AUTH-01** | The system shall allow authorized users to log in using valid credentials. | High |
| **FR-AUTH-02** | The system shall reject login attempts where the username or password is invalid. | High |
| **FR-AUTH-03** | The system shall store passwords using secure hashing rather than plain text. | High |
| **FR-AUTH-04** | The system shall provide logout functionality that ends the active user session. | High |
| **FR-AUTH-05** | The system shall restrict access to modules according to assigned user role. | High |
| **FR-AUTH-06** | The system shall allow an Administrator to create, update, deactivate and view user accounts. | Medium |
| **FR-AUTH-07** | The system shall prevent inactive users from logging in. | Medium |
| **FR-AUTH-08** | The system shall record important login and account management actions in the audit log. | Medium |

### 4.3 Campus, Department and Location Requirements

| Requirement ID | Requirement Statement | Priority |
|---|---|---|
| **FR-MST-01** | The system shall allow an Administrator to register campuses such as Nakawa and Namanve. | High |
| **FR-MST-02** | The system shall allow departments or sections to be linked to a campus. | High |
| **FR-MST-03** | The system shall allow inventory locations such as stores, laboratories, workshops, offices and rooms to be recorded. | High |
| **FR-MST-04** | The system shall allow inventory reports to be filtered by campus, department and location. | High |
| **FR-MST-05** | The system shall prevent deletion of a campus, department or location that is already linked to active inventory records. | Medium |

### 4.4 Inventory Catalogue and Asset Requirements

| Requirement ID | Requirement Statement | Priority |
|---|---|---|
| **FR-INV-01** | The system shall allow authorized users to register inventory items with item code, item name, category, description and unit of measure. | High |
| **FR-INV-02** | The system shall allow authorized users to classify items as consumable stock or fixed asset. | High |
| **FR-INV-03** | The system shall allow item records to store brand, model, specification, reorder level and optional image. | High |
| **FR-INV-04** | The system shall allow fixed assets to be registered as individual asset instances with serial number, asset tag, acquisition batch, purchase year, condition and location. | High |
| **FR-INV-05** | The system shall ensure that serial numbers and asset tags are unique where they are provided. | High |
| **FR-INV-06** | The system shall allow asset condition to be recorded as operational, faulty, under maintenance, retired, disposed or lost. | High |
| **FR-INV-07** | The system shall allow item records to be edited by authorized users while preserving transaction history. | High |
| **FR-INV-08** | The system shall allow deactivation of item records that should no longer be used for new transactions. | Medium |
| **FR-INV-09** | The system shall support search by item name, item code, brand, model, serial number, category, campus and department. | High |
| **FR-INV-10** | The system shall allow item images in approved formats such as JPG or PNG where image upload is enabled. | Low |

*Table 10 Inventory Catalogue and Asset Requirements*

## 4.5 Stock Management Requirements

| Requirement ID | Requirement Statement | Priority |
|---|---|---|
| **FR-STK-01** | The system shall allow the Store Manager to record stock received with item, quantity, supplier, date, receiving user and reference note. | High |
| **FR-STK-02** | The system shall allow the Store Manager to record stock issued with item, quantity, requesting department, receiving person, issue date and issuing user. | High |
| **FR-STK-03** | The system shall automatically update stock balance after each approved stock in or stock out transaction. | High |
| **FR-STK-04** | The system shall prevent stock issue where the requested quantity is greater than available quantity. | High |
| **FR-STK-05** | The system shall allow stock transfer between campus, department or location records where authorized. | High |
| **FR-STK-06** | The system shall allow stock return to stores and update the balance accordingly. | Medium |
| **FR-STK-07** | The system shall allow stock adjustment for approved correction of physical count differences. | Medium |
| **FR-STK-08** | The system shall maintain a transaction history for every stock movement. | High |
| **FR-STK-09** | The system shall show low stock alerts when available quantity falls below the reorder level. | High |
| **FR-STK-10** | The system shall support reports for stock received, stock issued, current stock and stock movement by date range. | High |

*Table 11 Stock Management Requirements*

## 4.6 Supplier Management Requirements

| Requirement ID | Requirement Statement | Priority |
|---|---|---|
| **FR-SUP-01** | The system shall allow authorized users to register supplier details including name, contact person, phone, email and address. | High |
| **FR-SUP-02** | The system shall allow supplier records to be updated by authorized users. | Medium |
| **FR-SUP-03** | The system shall link supplier records to stock received transactions. | High |
| **FR-SUP-04** | The system shall allow users to view supplier transaction history. | Medium |
| **FR-SUP-05** | The system shall prevent deletion of suppliers already linked to inventory transactions. | Medium |

*Table 12 Supplier Management Requirements*

## 4.7 Maintenance and Condition Tracking Requirements

| Requirement ID | Requirement Statement | Priority |
|---|---|---|
| **FR-MNT-01** | The system shall allow authorized users to record maintenance events for fixed assets. | Medium |
| **FR-MNT-02** | A maintenance record shall include asset, reported problem, action taken, maintenance date, responsible technician and resulting condition. | Medium |
| **FR-MNT-03** | The system shall allow reports of faulty assets by campus, department, category and condition. | High |
| **FR-MNT-04** | The system shall preserve historical condition changes for accountability. | Medium |

*Table 13 Maintenance and Condition Tracking Requirements*

## 4.8 Reporting and Dashboard Requirements

| Requirement ID | Requirement Statement | Priority |
|---|---|---|
| **FR-REP-01** | The system shall display dashboard summary cards for total inventory items, available stock, low stock items, faulty assets and recent transactions. | High |
| **FR-REP-02** | The system shall generate a report of computers by department, brand, specification, year of purchase, batch, serial number and operational status. | High |
| **FR-REP-03** | The system shall generate stock in and stock out statistics by month and year. | High |
| **FR-REP-04** | The system shall generate low stock reports by item category, campus and department. | High |
| **FR-REP-05** | The system shall generate supplier-based reports showing received items and dates. | Medium |
| **FR-REP-06** | The system shall allow report filtering by date range, category, campus, department, supplier and condition. | High |
| **FR-REP-07** | The system should provide simple graphical reports using charts where practical. | Medium |
| **FR-REP-08** | The system should support printing or export of key reports during demonstration. | Medium |

*Table 14 Reporting and Dashboard Requirements*

### 4.9 Security and Audit Requirements

| Requirement ID | Requirement Statement | Priority |
|---|---|---|
| **FR-SEC-01** | The system shall validate required fields before saving any record. | High |
| **FR-SEC-02** | The system shall use parameterized queries for database operations. | High |
| **FR-SEC-03** | The system shall restrict each page or action according to user role. | High |
| **FR-SEC-04** | The system shall record audit logs for login, item creation, item update, stock receipt, stock issue, transfer, adjustment and report access where applicable. | High |
| **FR-SEC-05** | An audit log shall include user, action, affected module, affected record, timestamp and device or session reference where practical. | Medium |
| **FR-SEC-06** | The system shall prevent unauthorized users from accessing protected pages through direct URL entry. | High |
| **FR-SEC-07** | The system shall sanitize user output to reduce cross site scripting risks. | High |
| **FR-SEC-08** | The system shall protect active sessions using server side session management. | High |

*Table 15 Security and Audit Requirements*

## 5.0 Non Functional Requirements

| Requirement ID | Category | Requirement Statement | Priority |
|---|---|---|---|
| **NFR-SEC-01** | Security | The system shall protect user accounts through hashed passwords, role restrictions and controlled sessions. | High |
| **NFR-SEC-02** | Security | The system shall reduce SQL injection risk through parameterized queries. | High |
| **NFR-PER-01** | Performance | Common searches and dashboard summaries should return within acceptable time during the prototype demonstration. | Medium |
| **NFR-REL-01** | Reliability | The system shall preserve stock transaction history after item updates. | High |
| **NFR-AVL-01** | Availability | The system should remain available during normal working hours when hosted on the local server. | Medium |
| **NFR-USE-01** | Usability | The interface shall use clear labels, simple forms and readable tables for staff users. | High |
| **NFR-MNT-01** | Maintainability | The source code should be organized into understandable modules for future maintenance. | Medium |
| **NFR-DAT-** | Data Integrity | The database shall enforce primary keys, foreign keys | High |

| | | | |
|---|---|---|---|
| **01** | | and required fields for important records. | |
| **NFR-SCA-01** | Scalability | The database shall support adding new campuses, departments, categories and locations without redesign. | High |
| **NFR-BCK-01** | Backup | The system should allow database backup through the approved database administration process. | Medium |
| **NFR-CMP-01** | Compatibility | The system shall run on a standard modern browser within the local development environment. | Medium |

*Table 16 Non-Functional Requirements*

## 6.0 Database Requirements

### 6.1 Core Data Requirements

| Proposed Table | Purpose |
|---|---|
| **roles** | Stores user role names such as Administrator, Store Manager, Staff User, Management Viewer and Auditor |
| **users** | Stores login and profile details for system users and links each user to a role |
| **campuses** | Stores campus records such as Nakawa and Namanve |
| **departments** | Stores departments or sections and links them to campuses |
| **locations** | Stores physical locations such as stores, laboratories, workshops, offices and rooms |
| **categories** | Stores item categories such as ICT Equipment, Laboratory Equipment, Office Assets and Consumables |
| **suppliers** | Stores supplier contact and identification details |
| **inventory_items** | Stores general item catalogue details such as item code, name, category, unit, description, brand and reorder level |
| **asset_instances** | Stores individual fixed asset records such as serial number, asset tag, purchase year, batch, condition and location |
| **stock_balances** | Stores available quantities per item, campus, department or location |
| **stock_transactions** | Stores stock in, stock out, transfer, return and adjustment records |
| **maintenance_records** | Stores maintenance and condition history for fixed assets |
| **audit_logs** | Stores important user actions for accountability |

*Table 17 Core Data Requirements*

### 6.2 Database Business Rules

1. Each user shall belong to one role.
2. Each department shall belong to one campus.
3. Each location shall belong to one campus and may belong to one department.
4. Each inventory item shall belong to one category.
5. Consumable items shall be tracked mainly by quantity and stock balance.
6. Fixed assets shall be tracked individually through asset instance records.
7. A stock transaction shall affect one inventory item and shall be recorded by one authorized user.
8. A stock received transaction may be linked to one supplier.
9. A fixed asset may have many maintenance records over time.
10. Audit logs shall not be edited through ordinary system forms.

### 6.3 Normalization Requirement

The database shall be normalized to at least Third Normal Form. This means each table shall represent one clear subject, repeating groups shall be removed, non key attributes shall depend on the full key and transitive dependencies shall be avoided. This is necessary because UIRI inventory data includes users, departments, suppliers, items, stock transactions and individual assets, which should not be stored in one flat spreadsheet style table [5].

## 7.0 Use Case Summary

| Use Case ID | Use Case Name | Primary Actor | Expected Result |
|---|---|---|---|
| **UC-01** | Log in to the system | Administrator, Store Manager, Staff User | User enters valid credentials and accesses permitted dashboard |
| **UC-02** | Manage user accounts | Administrator | Administrator creates, updates or deactivates user accounts |
| **UC-03** | Register inventory item | Administrator, Store Manager | Item details are saved in the inventory catalogue |
| **UC-04** | Register fixed asset instance | Store Manager | Asset with serial number, tag, batch and condition is recorded |
| **UC-05** | Record stock received | Store Manager | Stock quantity increases and transaction history is recorded |
| **UC-06** | Record stock issued | Store Manager | Stock quantity decreases and issue history is recorded |
| **UC-07** | Transfer stock or asset | Store Manager | Location or department assignment is updated with transaction record |
| **UC-08** | Record maintenance event | Store Manager, Technician where authorized | Fault or repair action is recorded for the selected asset |
| **UC-09** | Search inventory records | All authorized users | Search results are displayed according to access level |

| | | | |
|---|---|---|---|
| **UC-10** | Generate reports | Administrator, Store Manager, Management Viewer | Selected report is generated using filters |
| **UC-11** | Review audit logs | Administrator, Auditor | User activity history is displayed for accountability |

*Table 18 Use Case Summary*

## 8.0 Requirements Traceability Matrix

The traceability matrix links major requirements to the project objectives and expected test evidence. It will guide implementation and validation during the remaining project weeks.

| Objective | Objective Summary | Linked Requirements | Test Evidence |
|---|---|---|---|
| **Objective 1** | Study users, records, reports and business rules | FR-MST-01 to FR-MST-05, FR-REP-01 to FR-REP-08 | Approved SRS and requirements review notes |
| **Objective 2** | Design normalized relational database | Database Requirements 6.1 to 6.3 | Data dictionary, ERD and normalized schema |
| **Objective 3** | Implement secure authentication and access control | FR-AUTH-01 to FR-AUTH-08, NFR-SEC-01 | Login tests and role access tests |
| **Objective 4** | Enable inventory and asset tracking | FR-INV-01 to FR-INV-10 | Item records, asset records and search tests |
| **Objective 5** | Support stock movement and low stock alerts | FR-STK-01 to FR-STK-10 | Stock transaction tests and balance verification |
| **Objective 6** | Provide supplier management | FR-SUP-01 to FR-SUP-05 | Supplier records and transaction history tests |
| **Objective 7** | Generate dashboard and reports | FR-REP-01 to FR-REP-08 | Dashboard screenshot and sample reports |
| **Objective 8** | Enforce validation, password protection, sessions and audit logs | FR-SEC-01 to FR-SEC-08, NFR-DAT-01 | Security test notes and audit log entries |

*Table 19 Requirements Traceability Matrix*

## 9.0 Acceptance Criteria

The system shall be considered acceptable for the internship prototype when the main database and web application functions can be demonstrated using realistic test data and the results match the requirements defined in this document.

| Area | Acceptance Criterion |
|---|---|
| **Authentication** | At least one Administrator, one Store Manager and one Staff User can log in and access only permitted functions |
| **Inventory Registration** | Items and asset instances can be created, edited, searched and viewed |
| **Stock Transactions** | Stock in, stock out, transfer and return actions update balances correctly |
| **Low Stock Alerts** | Items below reorder level appear in the low stock list or dashboard |
| **Supplier Management** | Supplier records can be created and linked to stock received transactions |
| **Reports** | Dashboard and report filters show useful inventory summaries |
| **Audit Trail** | Important user actions are captured in the audit log |
| **Database Integrity** | Primary key and foreign key relationships prevent inconsistent records |
| **Documentation** | SRS, data dictionary, ERD, UML diagrams, user manual and final report are prepared |
| **Presentation** | The team can demonstrate the system and explain its database design decisions |

*Table 20 Acceptance Criteria Summary*

# 10.0 Appendices

## 10.1 Sample Inventory Reports Expected

- All computers in Mechatronics with brand, specification, serial number, year of purchase, batch and current condition.
- All faulty ICT equipment grouped by department and campus.
- Low stock consumables by category and location.
- Monthly stock received and stock issued report.
- Supplier transaction history by date range.
- Asset movement history showing transfers between departments or locations.

## 10.2 Next Design Documents

After approval of this SRS, the next deliverable shall be the Database Design document. That document shall contain the data dictionary, conceptual ERD, logical ERD, normalization evidence and the first set of Microsoft Visio diagrams. The SRS therefore acts as the official bridge between the proposal and database design.

# 11.0 References

[1] ICT Team, UIRI Namanve, "Internship Project Question: Design and Development of an Inventory Management System for Uganda Industrial Research Institute (UIRI)," Uganda Industrial Research Institute, Namanve, Uganda, Project Brief, 2026.

[2] K. P. Kato, K. Tracy and A. D. Kamate, "System Proposal: Design and Development of a Web Based Inventory Management System for Uganda Industrial Research Institute," Makerere University ICT Internship Project, 2026.

[3] Uganda Industrial Research Institute, "Uganda Industrial Research Institute: Home," UIRI Official Website. Accessed: Jun. 28, 2026. [Online]. Available: https://uiri.go.ug/

[4] Ministry of Trade, Industry and Cooperatives, "Uganda Industrial Research Institute," Government of Uganda. Accessed: Jun. 28, 2026. [Online]. Available: https://www.mtic.go.ug/uganda-industrial-research-institute/

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[6] IEEE Computer Society, "IEEE Recommended Practice for Software Requirements Specifications," IEEE Std 830-1998, 1998.